ORGNZ-99: Best Practice Summary & DQL Reference

> **Series:** ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 99 | **Created:** March 2026 | **Last Updated:** 10/14/2026

Overview

This notebook consolidates every actionable best practice from the ORGNZ series (notebooks 01-10) into a single definitive reference. Each section provides the exact setting, value, or action to take — and a **Validation Queries** subsection with executable DQL to verify your configuration against a live Grail environment.

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> Each section ends with a **Validation Queries** block — executable DQL to verify your configuration against a live Grail environment.

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1. Bucket Design

#	Best Practice	Recommended Setting/Value	Priority	Category
1	Keep per-bucket ingest under 1 TB/day	Target: <1 TB/day per bucket; split at >2 TB/day (Dynatrace guidance)	Critical	Performance
2	One data type per bucket	Each bucket stores exactly one table type (logs, metrics, spans, events, bizevents, security.events)	Critical	Architecture
3	Use consistent naming convention	Format: `<provider>_<type>_<retention>` or `<team>_<type>_<retention>` (e.g., `aws_logs_35d`, `team_platform_logs`)	Critical	Naming
4	Bucket names: lowercase, letter-first, underscores/hyphens only	Regex: `^[a-z][a-z0-9_-]*$`; never use `default_*` prefix	Critical	Naming
5	Plan bucket names before creation	Bucket names are immutable after creation; display names can be changed	Critical	Planning
6	Stay within the custom-bucket environment default — 250 from SaaS 1.346 (80 on earlier versions)	Budget buckets across teams; request increase only if justified	Recommended	Planning
7	Create dedicated buckets per cost center	One bucket per business unit/LOB for direct GiB/day cost attribution	Recommended	Cost
8	Create separate buckets for compliance data	Dedicated bucket (e.g., `compliance_audit_logs`) with extended retention for regulatory requirements	Recommended	Compliance
9	Create short-retention buckets for debug logs	Bucket: `debug_logs_7d` with 7-day retention for high-volume verbose logs	Recommended	Cost
10	Split high-volume buckets to maintain queryability	If bucket exceeds 1 TB/day, split by team, environment, or application	Recommended	Performance

Validation Queries — Bucket Design

List all buckets with data type, retention, and estimated storage:

```
// Inventory of all Grail buckets — verify data types, retention, and naming convention
fetch dt.system.buckets
| fields name, display_name, dt.system.table, retention_days, estimated_uncompressed_bytes
| sort dt.system.table asc, name asc
```

List all built-in buckets (use to compare your environment to the expected defaults):

```
// Discover all built-in Grail buckets — compare against expected defaults
fetch dt.system.buckets
| filter startsWith(name, "default_") or startsWith(name, "dt_")
| fields name, display_name, dt.system.table, retention_days
| sort dt.system.table asc, name asc
```

2. Retention Strategy

#	Best Practice	Recommended Setting/Value	Priority	Category
11	Set debug/development log retention to 3-7 days	Bucket retention: 7 days maximum for DEBUG-level logs	Critical	Cost
12	Set standard operational log retention to 35 days	Bucket retention: 35 days for application and infrastructure logs	Recommended	Operations
13	Set performance analysis retention to 60-90 days	Bucket retention: 90 days for logs/events used in quarterly trend analysis (metrics retention is fixed at ~15 months in `default_metrics`)	Recommended	Operations
14	Set audit/compliance log retention to 365-3657 days	Bucket retention: match regulatory requirement (SOX: 7 years = 2,555 days; HIPAA: 6 years = 2,190 days)	Critical	Compliance
15	Set span retention to 10-35 days	Bucket retention: 10 days for default APM; 35 days if long-running incident correlation needed	Recommended	Operations
16	Set security event retention to 90-365 days	Bucket retention: 365 days for security investigation timelines	Recommended	Security
17	Never lower retention without understanding data loss	Reducing retention immediately deletes data older than the new setting; this is irreversible	Critical	Operations

Validation Queries — Retention Strategy

Compare retention settings across all bucket types:

```
// Compare bucket retention configurations — verify each tier matches its compliance
requirement
fetch dt.system.buckets
| fields name, dt.system.table, retention_days
| sort retention_days desc
```

3. Data Routing

#	Best Practice	Recommended Setting/Value	Priority	Category
18	Route data to custom buckets via OpenPipeline	Configure Storage-stage **Bucket assignment** processors (first match wins) with DQL matchers on `loglevel`, `dt.host_group.id`, `log.source`, `dt.cost.*`	Critical	Architecture
19	Decide where unmatched records go	There is no `default:` route setting; data with no bucket assignment goes to the scope's default bucket (`default_logs` for logs) — check unmatched records in the pipeline preview	Critical	Reliability
20	Route DEBUG logs to short-retention buckets	Bucket assignment matcher `loglevel == "DEBUG"` → bucket `debug_logs_7d`	Recommended	Cost
21	Route audit logs to compliance buckets	Bucket assignment matcher `matchesPhrase(log.source, "audit")` → bucket `audit_logs_365d`	Recommended	Compliance
22	Data cannot be moved between buckets after ingestion	Route new data via OpenPipeline; old data stays in the original bucket until retention expires	Critical	Architecture

Validation Queries — Data Routing

Verify routing rules are directing data to the expected buckets:

```
// Audit log volume by bucket — verify routing rules are directing data correctly
fetch logs, from:-1h
| summarize recordCount = count(), by:{dt.system.bucket}
| sort recordCount desc
```

```
// Track log ingest trends by bucket over the last day — identify routing anomalies
fetch logs, from:-24h
| summarize recordCount = count(), by:{time_bucket = bin(timestamp, 1h), dt.system.bucket}
| sort time_bucket desc
```

4. IAM and Access Control

#	Best Practice	Recommended Setting/Value	Priority	Category
23	Always include `storage:buckets:read` in every data access policy	Without bucket read, users cannot query any data; always pair with table permissions	Critical	Security
24	Assign policies to groups, never to individual users	Create IAM groups per team/role; assign all policies at the group level	Critical	Governance
25	Use the two-step policy pattern	Step 1: `ALLOW storage:buckets:read WHERE ...`; Step 2: `ALLOW storage:logs:read` (or other table types)	Critical	Security
26	Use STARTSWITH for team bucket prefixes	Policy: `ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "team_platform_"` — automatically includes future buckets with matching prefix	Recommended	Scalability
27	Use MATCH operator for array fields	When `dt.security_context` holds an array, use `MATCH` (not `=`, `STARTSWITH`, or `IN` — those always return false for arrays)	Critical	Security
28	Start with least privilege	Grant minimum required access; expand only when justified	Critical	Security
29	Use policy boundaries for reusable conditions	Define regional/environmental restrictions once as a boundary; apply to multiple policies	Recommended	Governance
30	Limit policy boundaries to 10 restrictions per boundary	Maximum 10 conditions per boundary; create additional boundaries if more are needed	Recommended	Architecture
31	Use fieldsets for sensitive fields	Grant `storage:fieldsets:read` on specific fieldsets (e.g. `builtin-sensitive-spans`, or a custom fieldset for log fields) only where needed — users without it don't see those fields. There is no per-field `DENY` permission	Recommended	Security
32	Test policies with sample users before deployment	Create a test user in the target group and verify query results before rolling out to the team	Critical	Operations

#	Best Practice	Recommended Setting/Value	Priority	Category
33	Document every IAM policy	Include name, description, tags, target group, and purpose in a policy registry	Recommended	Governance

Validation Queries — IAM and Access Control

Verify bucket access and data distribution across teams:

```
// Verify bucket access — distribution of accessible log data by bucket
fetch logs, from:-1h
| summarize count = count(), by:{dt.system.bucket}
| sort count desc
```

```
// Check namespace-based access — verify namespace-scoped policies return expected data
fetch logs, from:-1h
| filter isNotNull(k8s.namespace.name)
| summarize count = count(), by:{k8s.namespace.name}
| sort count desc
```

```
// Check host group-based access — verify host-group-scoped policies return expected data
fetch logs, from:-1h
| filter isNotNull(dt.host_group.id)
| summarize count = count(), by:{dt.host_group.id}
| sort count desc
```

5. Security Context

#	Best Practice	Recommended Setting/Value	Priority	Category
34	Use `dt.security_context` for record-level access control	Set at the source first (host tag, `DT_TAGS`, K8s namespace labels, OTel resource attribute); OpenPipeline Set security context processor for computed values	Critical	Security
35	Use hierarchical encoding in security context values	Format: `<org>/<department>/<team>` (e.g., `acme/engineering/platform`); enables department-level access with `MATCH ('acme/engineering/*')`	Critical	Scalability
36	Use type prefixes for security context values	Prefix with `team:`, `lob:`, `env:`, `cloud:` (e.g., `team:checkout`, `lob:finance`, `env:production`)	Recommended	Naming

#	Best Practice	Recommended Setting/Value	Priority	Category
37	Plan security context hierarchy before implementation	Context values are assigned to records at ingest time; changing schema requires re-ingesting data	Critical	Planning
38	Use security context instead of additional buckets when at the bucket-count limit	Security context enables multi-team isolation within shared buckets without consuming bucket quota	Recommended	Scalability
39	Set security context on entities via Grail Security Context settings	Settings → Topology model → Grail Security Context; this scopes the entities only, not their logs/spans/metrics — use a host tag or primary-field enrichment for those	Recommended	Security
40	Always use MATCH for array-valued security context in IAM policies	`ALLOW storage:logs:read WHERE storage:dt.security_context MATCH ("team-a*")`	Critical	Security

Validation Queries — Security Context

Verify `dt.security_context` assignment, distribution, and hierarchical patterns:

```
// View logs with security context — confirm dt.security_context is being set
fetch logs, from:-1h
| filter isNotNull(dt.security_context)
| fields timestamp, content, dt.security_context
| limit 20
```

```
// Summarize log volume by security context — verify naming conventions and distribution
fetch logs, from:-1h
| filter isNotNull(dt.security_context)
| summarize recordCount = count(), by:{dt.security_context}
| sort recordCount desc
| limit 20
```

```
// Filter logs by specific security context — test access control boundary for a team
fetch logs, from:-1h
| filter dt.security_context == "team:platform"
| limit 50
```

```
// Check hierarchical security context — verify parent-level wildcard access works
fetch logs, from:-1h
| filter isNotNull(dt.security_context)
| filter startsWith(dt.security_context, "finance/")
| summarize count = count(), by:{dt.security_context}
| sort count desc
```

```
// Audit security context coverage — percentage of logs with dt.security_context assigned
fetch logs, from:-1h
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context)),
    withoutContext = countIf(isNull(dt.security_context))
| fieldsAdd coverage = round(toDouble(withContext) / toDouble(total) * 100, decimals: 2)
```

6. Segment Design

#	Best Practice	Recommended Setting/Value	Priority	Category
41	Use segments for dynamic data views, not access control	Segments filter at query time; they do not enforce security boundaries — IAM policies do	Critical	Architecture
42	Stay within the 20-include limit per segment	Maximum 20 include blocks per segment; plan rule complexity accordingly	Critical	Architecture
43	One include block per data type	Cannot have two `logs` includes; combine conditions with `OR` within a single include	Critical	Architecture
44	Use Primary Grail Fields in segment filters	Filter on `dt.host_group.id`, `k8s.namespace.name`, `k8s.cluster.name`, `aws.account.id`, `azure.subscription`, `gcp.project.id` — these are indexed and propagate across all signal types	Critical	Performance
45	Use prefix-based naming for cross-signal filtering	Name host groups, namespaces, etc. with prefixes (e.g., `prod-web-tier` not `web-tier-prod`) so a prefix (`= "prod-*"`) works on signal includes and on classic `entity.name` (the only classic-entity property that accepts starts-with)	Recommended	Naming
46	Create one segment per dimension	Separate segments for platform, application, environment, and region — users combine them as needed	Recommended	Architecture
47	Add event includes for detected problem filtering	Define events include with `event.kind = "DAVIS_PROBLEM"` — entity includes alone do not filter problems	Critical	Troubleshooting

#	Best Practice	Recommended Setting/Value	Priority	Category
48	Set official segments to Public visibility	Platform-team segments: *Anyone in the environment*; application-team segments: *Unlisted*, surfaced through the team's shared notebooks and dashboards (segments have no share-with-group setting)	Recommended	Governance
49	Use consistent segment naming prefixes	Prefixes: `team-`, `env-`, `app-`, `region-` (e.g., `team-platform-infra`, `env-production`)	Recommended	Naming
50	Test segment filter conditions with DQL before creating the segment	Run the equivalent `filter` clause as a DQL query; verify record counts match expectations	Critical	Operations
51	Limit segment variable dropdowns to 10,000 values	Maximum 10,000 values per variable; 100 selected values per segment	Recommended	Architecture
52	Use entity relationship traversal for comprehensive segments	Top-down: host → process group → service; bottom-up: service → process group → host	Recommended	Architecture
53	Segments reduce scanned bytes	Applying a segment injects filters at query time, reducing data scanned and improving performance on high-volume buckets	Recommended	Performance

Validation Queries — Segment Design

Simulate segment filter conditions before committing to a segment definition:

```
// Simulate host-group-based segment filter on logs
fetch logs, from:-1h
| filter isNotNull(dt.host_group.id)
| summarize count = count(), by:{dt.host_group.id}
| sort count desc
| limit 10
```

```
// Simulate a primary-tag segment filter on the service list: what a Data (all types) include sees.
// Service metrics carry primary Grail tags, not classic entity tags. Replace primary_tags.app
// with your key. Validated 10/14/2026 with a tenant-specific key: 3 values, 4-17 services each.
timeseries requests = sum(dt.service.request.count), from:-2h, by:{primary_tags.app, dt.smartscape.service}
| filter isNotNull(primary_tags.app)
| fieldsAdd total = arraySum(requests)
| summarize {services = countDistinct(dt.smartscape.service), requests = sum(total)}, by:{primary_tags.app}
| sort requests desc
```

```
// Simulate combined segment filter — verify host group + namespace combination returns expected logs
fetch logs, from:-1h
| filter isNotNull(dt.host_group.id) and isNotNull(k8s.namespace.name)
| summarize count = count(), by:{dt.host_group.id, k8s.namespace.name}
| sort count desc
| limit 20
```

7. Metadata Enrichment

#	Best Practice	Recommended Setting/Value	Priority	Category
54	Enrich all signals with Primary Grail Fields before building segments	Verify `dt.host_group.id`, `k8s.namespace.name`, `k8s.cluster.name` are present on logs, spans, and metrics	Critical	Architecture
55	Set `dt.security_context` via a host tag for dedicated infrastructure	Host tag `dt.security_context=<team>` (installation, `oneagentctl --set-host-tag`, or Deployment Status)	Recommended	Security
56	Set `dt.cost.costcenter` and `dt.cost.product` on hosts	Host tags: `dt.cost.costcenter=<center>`, `dt.cost.product=<product>` — these propagate to service metrics	Recommended	Cost
57	Use `DT_TAGS` environment variable for shared infrastructure	Set `DT_TAGS=primary_tags.team=<team>` per process when multiple apps share the same host	Recommended	Architecture

#	Best Practice	Recommended Setting/Value	Priority	Category
58	Restart applications after enrichment changes for spans	Logs: no restart needed (OneAgent auto-enriches); Spans: application restart required; Metrics: OneAgent restart applies automatically	Critical	Operations
59	Verify enrichment coverage before building segments	Run DQL audit query to check field coverage percentage across signal types; fields with 0% coverage need enrichment	Critical	Operations
60	Use Primary Grail Tags (`primary_tags.*`) when standard fields do not match your organization	Set via **host tags** (`hostautotag.conf` / `oneagentctl --set-host-tag`) or the `DT_TAGS` environment variable; propagates across logs, spans, and topology. **Host properties no longer work for this** — see row 60a	Recommended	Architecture
60a	Do not set host-level `primary_tags.*` via `hostcustomproperties.conf`	**Breaking, OneAgent 1.345:** *"the OS Agent derives `primary_tags` from host tags (`hostautotag.conf`) rather than from host properties (`hostcustomproperties.conf`) at the host level."* Existing primary tags configured through host properties silently stop being promoted once a host reaches 1.345 — audit before the fleet upgrades, and migrate them to host tags	Critical	Migration

Validation Queries — Metadata Enrichment

Audit primary Grail field coverage before building segment definitions:

```
// Audit primary Grail field coverage on logs — low-coverage fields need enrichment before segment use
fetch logs, from:-1h
| summarize
    total = count(),
    with_namespace = countIf(isNotNull(k8s.namespace.name)),
    with_cluster = countIf(isNotNull(k8s.cluster.name)),
    with_host_group = countIf(isNotNull(dt.host_group.id)),
    with_security_ctx = countIf(isNotNull(dt.security_context))
| fieldsAdd
    ns_pct = round(toDouble(with_namespace) / toDouble(total) * 100, decimals: 1),
    cluster_pct = round(toDouble(with_cluster) / toDouble(total) * 100, decimals: 1),
    hg_pct = round(toDouble(with_host_group) / toDouble(total) * 100, decimals: 1),
    sec_pct = round(toDouble(with_security_ctx) / toDouble(total) * 100, decimals: 1)
```

```
// Audit primary Grail field coverage on spans — spans require app restart to pick up new enrichment
fetch spans, from:-1h
| summarize
    total = count(),
    with_namespace = countIf(isNotNull(k8s.namespace.name)),
    with_cluster = countIf(isNotNull(k8s.cluster.name)),
    with_host_group = countIf(isNotNull(dt.host_group.id)),
    with_service = countIf(isNotNull(dt.smartscape.service))
| fieldsAdd
    ns_pct = round(toDouble(with_namespace) / toDouble(total) * 100, decimals: 1),
    cluster_pct = round(toDouble(with_cluster) / toDouble(total) * 100, decimals: 1),
    hg_pct = round(toDouble(with_host_group) / toDouble(total) * 100, decimals: 1),
    svc_pct = round(toDouble(with_service) / toDouble(total) * 100, decimals: 1)
```

```
// Variable DQL: list host groups. Use as a segment variable for host-group-scoped filters.
// There is no HOST_GROUP Smartscape node type; the host group is the dt.host_group.id field
// on HOST nodes. Validated 10/14/2026; 0 bytes scanned.
smartscapeNodes "HOST"
| filter isNotNull(dt.host_group.id)
| fields host_group = dt.host_group.id
| dedup host_group
| sort host_group asc
```

```
// Variable DQL: list Kubernetes namespaces. Use as a segment variable for namespace-scoped filters.
// Validated 10/14/2026: 12 namespace names across 23 K8S_NAMESPACE nodes; 0 bytes scanned.
smartscapeNodes "K8S_NAMESPACE"
| fields namespace = name
| dedup namespace
| sort namespace asc
```

8. Query Performance

#	Best Practice	Recommended Setting/Value	Priority	Category
61	Understand the default 500 GB scan limit	`fetch` stops at 500 GB scanned by default (`scanLimitGBytes`); at 1 TB/day ingest an unmodified query reaches back ~12 hours — raise `scanLimitGBytes` or narrow the query	Critical	Performance
62	Target specific buckets in queries	Use `fetch logs, bucket:{"team_logs", "app_logs_*"}` to avoid scanning all buckets	Recommended	Performance
63	Use exact match (`=`) over pattern match in segment filters	`=` is faster than `starts-with` ; use exact values when known	Recommended	Performance
64	Minimize segment filter expressions	Fewer conditions = better query performance; maximum 10 expressions per filter	Recommended	Performance
65	Use OpenPipeline pre-processing for complex segment logic	If segment conditions would be complex, pre-enrich a simple field (e.g., `dt.security_context`) and filter on that	Recommended	Performance
66	Maximum 1,000 records returned per query	Use aggregations (`summarize` , `makeTimeseries`) for larger datasets	Recommended	Performance
67	Maximum 10 segments per query	Do not stack more than 10 segments; design broader segments instead of many narrow ones	Recommended	Architecture

Validation Queries — Query Performance

Use bucket targeting to control scan scope and stay within the 500 GB scan limit:

```
// Query from multiple buckets simultaneously — reduce scan scope by naming targets explicitly
// Bucket names that don't exist are skipped with no warning — a misspelled name looks like an empty bucket.
// Check names first: fetch dt.system.buckets | filter dt.system.table == "logs" | fields name
fetch logs, from:-1h, bucket:{"default_logs", "audit_logs", "security_logs"}
| limit 100
```

```
// Filter by bucket within query — useful when bucket list is dynamic
fetch logs, from:-1h
| filter dt.system.bucket == "default_logs"
// status == "ERROR" also counts SEVERE, CRITICAL and FATAL logs; loglevel == "ERROR"
misses them
| filter status == "ERROR"
| limit 50
```

9. Governance and Operations

#	Best Practice	Recommended Setting/Value	Priority	Category
68	Schedule regular access reviews	Quarterly review of all IAM policies and group memberships; remove unnecessary permissions	Critical	Governance
69	Document all bucket purposes	Maintain a registry mapping each bucket to its owner, data type, retention, cost center, and routing rule	Recommended	Governance
70	Document all security context values	Maintain a registry of all `dt.security_context` values, their meaning, and which IAM policies reference them	Recommended	Governance
71	Platform team owns infrastructure segments	Platform/SRE team creates and maintains public environment and infrastructure segments	Recommended	Governance
72	Application teams own app-specific segments	Each team creates segments for its applications, kept *Unlisted* and referenced from the team's shared documents	Recommended	Governance
73	Review segments quarterly	Remove unused segments; verify filter conditions still match data; check for empty results	Recommended	Operations
74	Use combined approach for defense in depth	Buckets for retention/cost + Security context for access control + Segments for dynamic views — all three together	Critical	Architecture
75	Implement in phases	Phase 1: Plan → Phase 2: Buckets + routing → Phase 3: IAM policies → Phase 4: Segments → Phase 5: Governance	Recommended	Operations
76	Selective record deletion is not possible	Cannot delete individual records; wait for bucket retention to expire, or truncate the entire bucket	Critical	Operations

#	Best Practice	Recommended Setting/Value	Priority	Category
77	Bucket deletion destroys all data permanently	Deleting a bucket removes ALL records immediately and irreversibly	Critical	Operations

Validation Queries — Governance and Operations

Routine audit queries for operational health checks:

```
// Audit bucket data distribution — verify all expected buckets are receiving data
fetch logs, from:-1h
| summarize
    recordCount = count(),
    by:{dt.system.bucket}
| sort recordCount desc
```

```
// Audit bucket retention settings — verify retention tiers match compliance requirements
fetch dt.system.buckets
| fields name, dt.system.table, retention_days
| sort dt.system.table asc, retention_days desc
```

Audit bucket management events — who created, updated, truncated, or deleted a bucket and when:

```
// Audit all bucket management actions — create, update, truncate, delete
// Bucket changes are rare: widen to from:-365d for a full year of history
fetch dt.system.events, from:-90d
| filter event.kind == "AUDIT_EVENT" and event.category == "BUCKET_MANAGEMENT"
| fields timestamp, event.type, resource, user.id, event.outcome
| sort timestamp desc
```

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